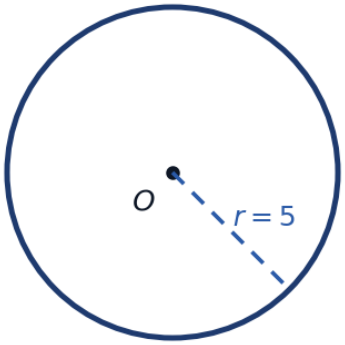
Trigonometric Functions and Circular Measure

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

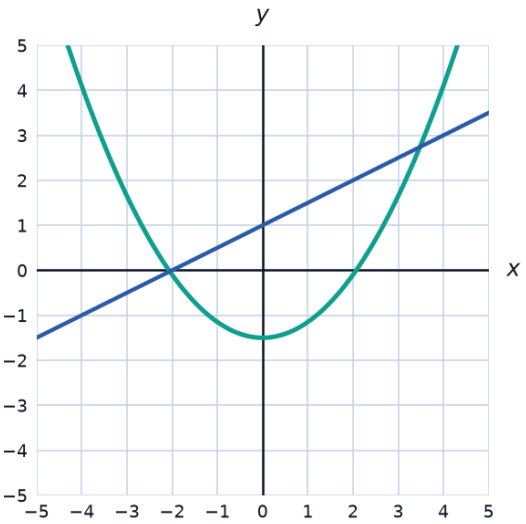
1 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

The diagram shows a sector of a circle with centre O. The arc length of the sector is given by 6π cm and the area of the sector is 27π cm^2. The radius of the circle is r cm and the angle of the sector is θ radians. Find the value of r and θ.



2 [Foundational] • Target: Radian Measure, Arc Length and Sectors

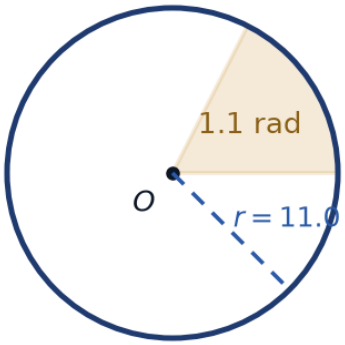
The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is 6 cm and the radius is 3 cm, find the angle θ in radians.



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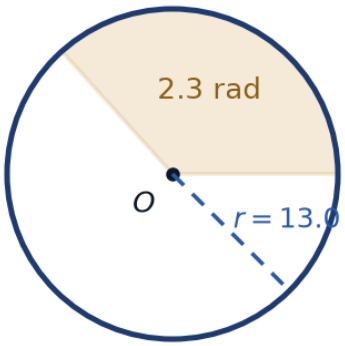
3 [Foundational] • Target: Trigonometric Identities and Ratios

A sector has radius 11 cm and angle 1.1 rad. Find the arc length.



4 [Foundational] • Target: Solving Trigonometric Equations

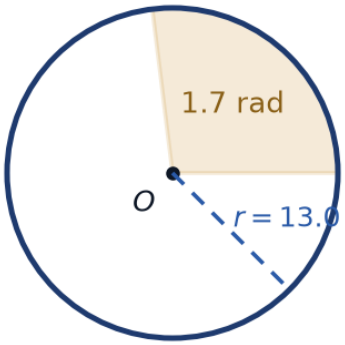
A sector has radius 13 cm and angle 2.3 rad. Find the arc length.



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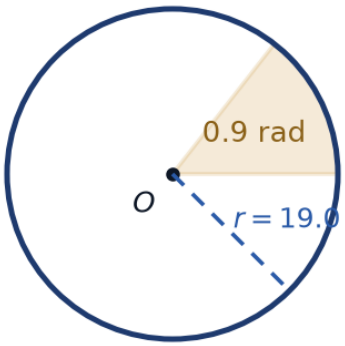
5 [Foundational] • Target: Double Angle Formulae and Graph Amplitude

A sector has radius 13 cm and angle 1.7 rad. Find the arc length.



6 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

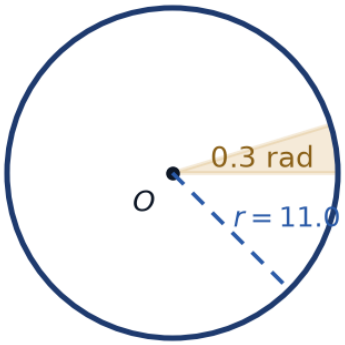
A sector has radius 19 cm and angle 0.9 rad. Find the arc length.



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7 [Foundational] • Target: Radian Measure, Arc Length and Sectors

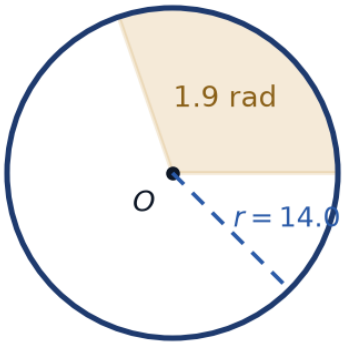
A sector has radius 11 cm and angle 0.3 rad. Find the arc length.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Trigonometric Identities and Ratios

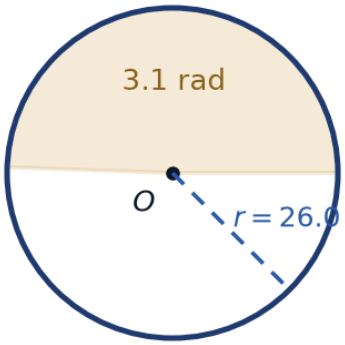
A sector has radius 14 cm and angle 1.9 rad. Find the arc length.



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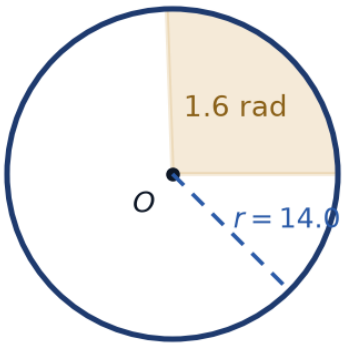
9 [Intermediate] • Target: Solving Trigonometric Equations

A sector has radius 26 cm and angle 3.1 rad. Find the arc length.



10 [Intermediate] • Target: Double Angle Formulae and Graph Amplitude

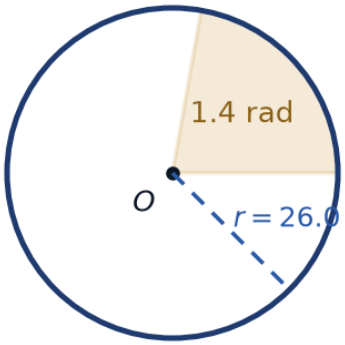
A sector has radius 14 cm and angle 1.6 rad. Find the arc length.



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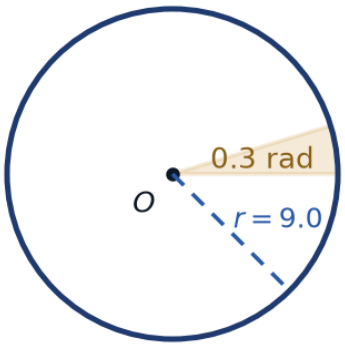
11 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector has radius 26 cm and angle 1.4 rad. Find the arc length.



12 [Intermediate] • Target: Radian Measure, Arc Length and Sectors

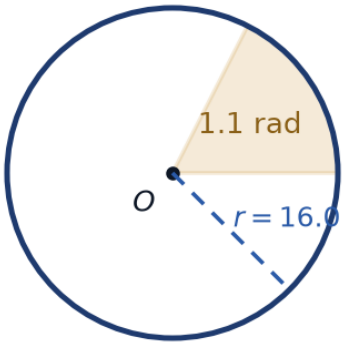
A sector has radius 9 cm and angle 0.3 rad. Find the arc length.



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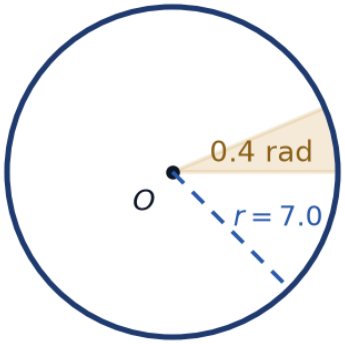
13 [Intermediate] • Target: Trigonometric Identities and Ratios

A sector has radius 16 cm and angle 1.1 rad. Find the arc length.



14 [Intermediate] • Target: Solving Trigonometric Equations

A sector has radius 7 cm and angle 0.4 rad. Find the arc length.

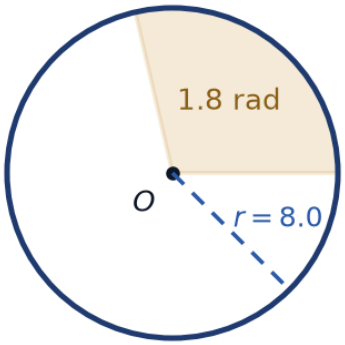


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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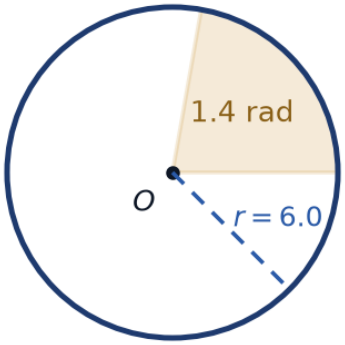
15 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 6.01 Permutations and Combinations

A sector has radius 8 cm and angle 1.8 rad. Find the arc length.



16 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

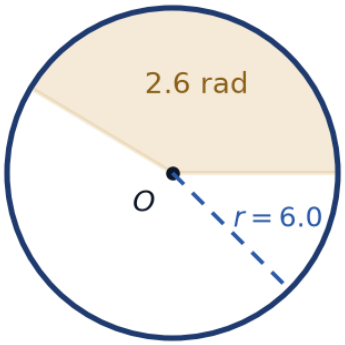
A sector has radius 6 cm and angle 1.4 rad. Find the arc length.



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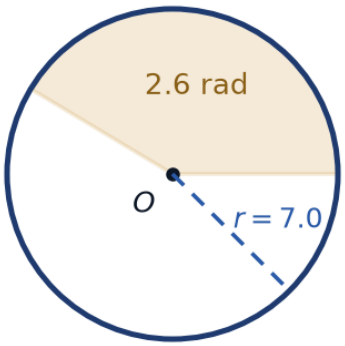
17 [Challenging] • Target: Radian Measure, Arc Length and Sectors • Cross-Lesson: 8.01 Integration and Kinematics

A sector has radius 6 cm and angle 2.6 rad. Find the arc length.



18 [Challenging] • Target: Trigonometric Identities and Ratios • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

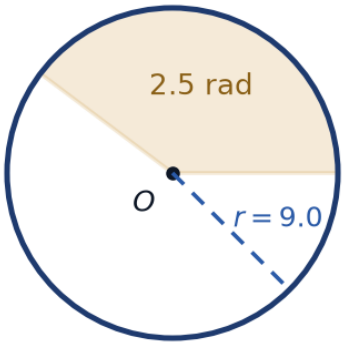
A sector has radius 7 cm and angle 2.6 rad. Find the arc length.



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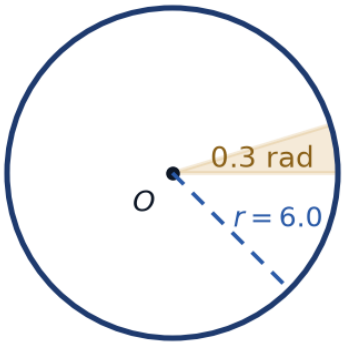
19 [Challenging] • Target: Solving Trigonometric Equations • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

A sector has radius 9 cm and angle 2.5 rad. Find the arc length.



20 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 3.01 Indices, Surds and Polynomials

A sector has radius 6 cm and angle 0.3 rad. Find the arc length.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius $r = 6 \text{ cm}$ and central angle $\theta = \frac{\pi}{3} \text{ rad}$. What is the arc length of the sector?

a. $\frac{6\pi}{3} \text{ cm}$

b. $\frac{36\pi}{6} \text{ cm}$

c. $6\pi \text{ cm}$

d. $\frac{6\pi}{6} \text{ cm}$

22 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius $r = 8 \text{ cm}$ and central angle $\theta = \frac{\pi}{4} \text{ rad}$. What is the arc length of the sector?

a. $\frac{8\pi}{4} \text{ cm}$

b. $\frac{64\pi}{8} \text{ cm}$

c. $8\pi \text{ cm}$

d. $\frac{8\pi}{8} \text{ cm}$

23 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius $r = 10 \text{ cm}$ and central angle $\theta = \frac{\pi}{5} \text{ rad}$. What is the arc length of the sector?

a. $\frac{10\pi}{5} \text{ cm}$

b. $\frac{100\pi}{10} \text{ cm}$

c. $10\pi \text{ cm}$

d. $\frac{10\pi}{10} \text{ cm}$

24 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius $r = 12 \text{ cm}$ and central angle $\theta = \frac{\pi}{6} \text{ rad}$. What is the arc length of the sector?

a. $\frac{12\pi}{6} \text{ cm}$

b. $\frac{144\pi}{12} \text{ cm}$

c. $12\pi \text{ cm}$

d. $\frac{12\pi}{12} \text{ cm}$

25 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius $r = 14 \text{ cm}$ and central angle $\theta = \frac{\pi}{7} \text{ rad}$. What is the arc length of the sector?

a. $\frac{196\pi}{14} \text{ cm}$

b. $\frac{14\pi}{7} \text{ cm}$

c. $14\pi \text{ cm}$

d. $\frac{14\pi}{14} \text{ cm}$

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26 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius \(r = 16 \text{ cm} \) and central angle \(\theta = \frac{\pi}{8} \text{ rad} \). What is the arc length of the sector?

a. \(\frac{16\pi}{8} \text{ cm} \)

b. \(\frac{256\pi}{16} \text{ cm} \)

c. \(16\pi \text{ cm} \)

d. \(\frac{16\pi}{16} \text{ cm} \)

27 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius \(r = 18 \text{ cm} \) and central angle \(\theta = \frac{\pi}{9} \text{ rad} \). What is the arc length of the sector?

a. \(\frac{18\pi}{9} \text{ cm} \)

b. \(\frac{324\pi}{18} \text{ cm} \)

c. \(18\pi \text{ cm} \)

d. \(\frac{18\pi}{18} \text{ cm} \)

28 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius \(r = 20 \text{ cm} \) and central angle \(\theta = \frac{\pi}{10} \text{ rad} \). What is the arc length of the sector?

a. \(20\pi \text{ cm} \)

b. \(\frac{400\pi}{20} \text{ cm} \)

c. \(\frac{20\pi}{10} \text{ cm} \)

d. \(\frac{20\pi}{20} \text{ cm} \)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Radian Measure, Arc Length and Sectors

A circle of radius 8 cm has a central sector with angle $\theta = \frac{3\pi}{4}$ rad. Find the exact arc length of the sector.

a. 6π cm

b. 12π cm

c. 4π cm

d. 24π cm

30 [Intermediate] • Target: Radian Measure, Arc Length and Sectors

Find the area of a sector of radius 6 cm having perimeter 22 cm.

a. 30 cm^2

b. 36 cm^2

c. 24 cm^2

d. 18 cm^2

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31 [Intermediate] • Target: Radian Measure, Arc Length and Sectors

A chord of length 10 cm subtends an angle of 2θ at the center of a circle of radius 13 cm. Find $\sin\theta$.

a. $\frac{5}{13}$

b. $\frac{12}{13}$

c. $\frac{10}{13}$

d. $\frac{5}{12}$

32 [Intermediate] • Target: Trigonometric Identities and Ratios

Simplify the trigonometric expression $\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}$.

a. $\sec\theta\csc\theta$

b. $\tan\theta$

c. 1

d. $\sec^2\theta$

33 [Intermediate] • Target: Trigonometric Identities and Ratios

Given that $\tan\theta = -\frac{3}{4}$ and θ lies in the second quadrant, find the value of $\cos\theta$.

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $\frac{3}{5}$

34 [Intermediate] • Target: Trigonometric Identities and Ratios

Prove the identity $\sec^2\theta - \tan^2\theta = 1$. Which of the following equals $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$?

a. 1

b. $\sec^2\theta$

c. $2\tan^2\theta$

d. 0

35 [Intermediate] • Target: Solving Trigonometric Equations

Solve $2\cos(2x) = 1$ for $0 \le x \le \pi$.

a. $x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$

b. $x = \frac{\pi}{3}$ and $x = \frac{2\pi}{3}$

c. $x = \frac{\pi}{6}$ only

d. $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$

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36 [Intermediate] • Target: Solving Trigonometric Equations

Find all solutions of $3\sin x = 2\cos^2 x$ in the interval $0^\circ \le x \le 360^\circ$.

a. 30° and 150°

b. 30° and 210°

c. 60° and 120°

d. 45° and 135°

37 [Intermediate] • Target: Solving Trigonometric Equations

Solve the equation $\tan(x + 45^\circ) = 1$ for $0^\circ \le x \le 360^\circ$.

a. $x = 0^\circ, 180^\circ, 360^\circ$

b. $x = 45^\circ, 225^\circ$

c. $x = 90^\circ, 270^\circ$

d. $x = 0^\circ$ only

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

If $\sin\theta = \frac{3}{5}$ and θ is acute, find the exact value of $\sin(2\theta)$.

a. $\frac{24}{25}$

b. $\frac{12}{25}$

c. $\frac{7}{25}$

d. $\frac{6}{5}$

39 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

State the amplitude and period of the function $y = 3\sin(4x) - 2$.

a. Amplitude $= 3$, Period $= \frac{\pi}{2}$

b. Amplitude $= 3$, Period $= \frac{\pi}{4}$

c. Amplitude $= 1$, Period $= 2\pi$

d. Amplitude $= 4$, Period $= \frac{\pi}{2}$

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40 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Express $\cos(2\theta)$ purely in terms of $\sin\theta$.

a. $1-2\sin^2\theta$

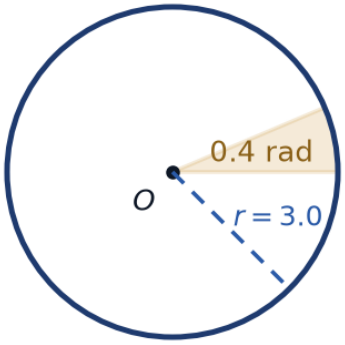
b. $2\sin^2\theta-1$

c. $1-\sin^2\theta$

d. $\sin^2\theta-1$

41 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A sector has radius 3 cm and angle 0.4 rad. Find the arc length.



a. 17/5

b. 3/5

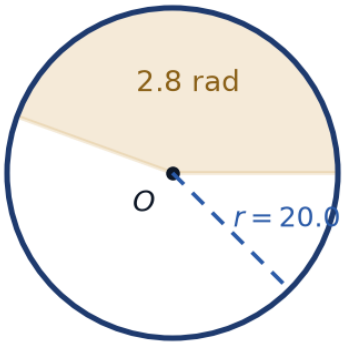
c. 11/5

d. 6/5

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42 [Challenging] • Target: Radian Measure, Arc Length and Sectors • Cross-Lesson: 6.01 Permutations and Combinations

A sector has radius 20 cm and angle 2.8 rad. Find the arc length.



a. 114/5

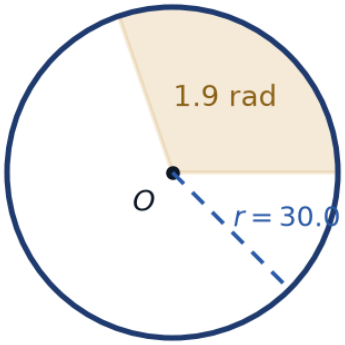
b. 56

c. 560

d. 28

43 [Challenging] • Target: Trigonometric Identities and Ratios • Cross-Lesson: 7.01 Differentiation and Rates of Change

A sector has radius 30 cm and angle 1.9 rad. Find the arc length.



a. 57

b. 855

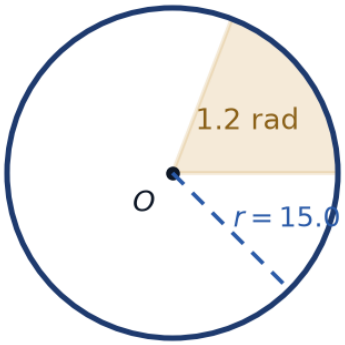
c. 58

d. 57/2

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44 [Challenging] • Target: Solving Trigonometric Equations • Cross-Lesson: 8.01 Integration and Kinematics

A sector has radius 15 cm and angle 1.2 rad. Find the arc length.



a. 9

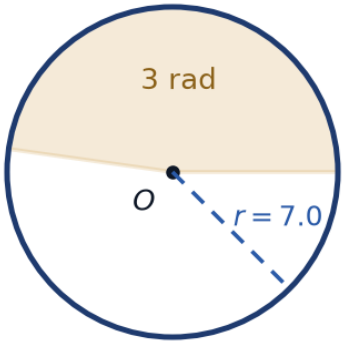
b. 81/5

c. 19

d. 18

45 [Challenging] • Target: Double Angle Formulae and Graph Amplitude • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

A sector has radius 7 cm and angle 3.0 rad. Find the arc length.



a. 21/2

b. 147/2

c. 10

d. 21

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